Weak Solutions and Characteristics

MA5565

The Wave Equation on $\mathbb{R}$

We consider the Cauchy problem on $\mathbb{R}$:

$$u_{tt} - c^2 u_{xx} = 0, \quad x \in \mathbb{R}, \; t > 0$$

with initial data

$$u(x,0) = f(x), \quad u_t(x,0) = g(x).$$

Classical solution assumption:

$$u \in C^2(\mathbb{R} \times [0,\infty)).$$

We now relax regularity and derive the weak formulation.

Weak Formulation: Test Functions

Let $\varphi \in C_c^\infty(\mathbb{R} \times [0, \infty))$.
Multiply the PDE by φ and integrate:

$$\int_0^\infty \int_{\mathbb{R}} (u_{tt} - c^2 u_{xx})\varphi \, dxdt = 0.$$

Integrate by parts in t and x. Boundary terms vanish because φ has compact support.

Weak Form of the Wave Equation

After integration by parts:

$$\int_0^\infty \int_{\mathbb{R}} \left(-u_t\varphi_t + c^2 u_x \varphi_x\right) dxdt$$

$$+ \int_{\mathbb{R}} g(x)\varphi(x,0)\, dx - \int_{\mathbb{R}} f(x)\varphi_t(x,0)\, dx = 0.$$

Definition: u is a weak solution if this identity holds for all $\varphi \in C_c^\infty$.

Energy Interpretation

Formal energy:

$$E(t) = \frac{1}{2}\int_{\mathbb{R}} \left(u_t^2 + c^2 u_x^2\right) dx.$$

For smooth solutions:

$$\frac{d}{dt}E(t) = 0.$$

Example: Constant Coefficients

Transport equation:

$$u_t + au_x = 0.$$

Rewrite as:

$$\frac{d}{dt}u(x(t), t) = 0$$

provided

$$\frac{dx}{dt} = a.$$

Solution:

$$u(x, t) = f(x - at).$$

Inhomogeneous Example

$$u_t + au_x = h(x,t).$$

Along characteristics $x(t) = x_0 + at$:

$$\frac{d}{dt}u(x(t),t) = h(x(t),t).$$

Integrate:

$$u(x,t) = f(x-at) + \int_0^t h(x-a(t-s),s)\,ds.$$

Definition: Characteristic Curves (Section 3.2)

For

$$a(x,t)u_x + b(x,t)u_t = c(x,t),$$

Characteristic curves satisfy:

$$\frac{dx}{ds} = a(x,t), \qquad \frac{dt}{ds} = b(x,t).$$

Along these curves:

$$\frac{d}{ds}u(x(s),t(s)) = c(x,t).$$

Geometric Interpretation

- Characteristics are curves tangent to (a, b).
- The PDE prescribes directional derivatives.
- Along characteristics, the PDE reduces to an ODE.
- Information propagates along these curves.

Initial Value Problem

Given data on a curve Γ:

$$u|_\Gamma = g.$$

Well-posed if:

- Γ is not characteristic.
- Characteristics intersect Γ transversely.

Characteristic curves determine domain of dependence.

Reminder: Burgers' Equation

We studied:

$$u_t + uu_x = 0.$$

This is nonlinear transport.

Characteristics for Burgers

Characteristic equations:

$$\frac{dx}{dt} = u, \qquad \frac{du}{dt} = 0.$$

Thus:

$$u = \text{constant along characteristics.}$$

Curves:

$$x(t) = x_0 + u_0(x_0)t.$$

Interpretation for Burgers

- Speed depends on the solution.
- Faster characteristics overtake slower ones.
- Intersection $\Rightarrow$ shock formation.
- Classical solution breaks down.

Motivation for weak solutions.

Big Picture

- Weak formulation: prepares for low regularity.
- First-order PDEs: solved via characteristics.
- Nonlinearity causes characteristic crossing.
- Leads naturally to weak and *entropy* solutions.

Section 3.1: Nonlinear First-Order Models

Model equations from mechanics etc.:

- Burgers equation
- Wave equation in first-order form
- The p-system
- Gas dynamics (Lagrangian)
- Gas dynamics (Eulerian)

These motivate characteristic curves and wave propagation.

Example 3.1: Burgers Equation

$$u_t + uu_x = 0.$$

Nonlinear transport.
Characteristic equations:

$$\frac{dx}{dt} = u, \qquad \frac{du}{dt} = 0.$$

Thus:

$$u = u_0(x_0), \qquad x(t) = x_0 + u_0(x_0)t.$$

Key feature: Characteristics depend on the solution itself.

Shock Formation in Burgers

- If $u_0'(x_0) < 0$, characteristics intersect.
- Classical solution breaks down.
- Gradient blows up in finite time.
- Motivates weak solutions.

Burgers is the prototype nonlinear conservation law.

Example 3.2: Wave Equation in u-v Form

Start from

$$\phi_{tt} - c^2\phi_{xx} = 0.$$

Introduce:

$$u = \phi_t, \qquad v = \phi_x.$$

System:

$$u_t - v_x = 0,$$

$$v_t - u_x = 0.$$

First-order linear hyperbolic system.

Characteristic Structure (Wave System)

Write in matrix form:

$$\begin{pmatrix} u \\ v \end{pmatrix}_t + \begin{pmatrix} 0 & -c^2 \\ -1 & 0 \end{pmatrix} \begin{pmatrix} u \\ v \end{pmatrix}_x = 0.$$

Eigenvalues:

$$\lambda = \pm c.$$

Thus:

$$x \mp ct = \text{constant}.$$

Two families of characteristics.

Example 3.3: The p-System

$$v_t - u_x = 0,$$
$$u_t + p(v)_x = 0.$$

Here:

- v = specific volume
- u = velocity
- $p(v)$ = pressure law

Matrix form:

$$\begin{pmatrix} v \\ u \end{pmatrix}_t + \begin{pmatrix} 0 & -1 \\ p'(v) & 0 \end{pmatrix} \begin{pmatrix} v \\ u \end{pmatrix}_x = 0.$$

Characteristic Speeds for the p-System

Eigenvalues satisfy:

$$\lambda^2 = p'(v).$$

Thus:

$$\lambda = \pm\sqrt{p'(v)}.$$

Interpretation:

- Nonlinear wave speeds.
- Depend on state v.
- Generalization of linear wave equation.

Example 3.4: Gas Dynamics (Lagrangian)

In Lagrangian coordinates:

$$v_t - u_x = 0,$$

$$u_t + p(v)_x = 0.$$

Same structure as p-system.
Nonlinear hyperbolic system.
Wave speeds:

$$\pm\sqrt{p'(v)}.$$

Example 3.5: Gas Dynamics (Eulerian)

Compressible Euler equations (1D):

$$\rho_t + (\rho u)_x = 0,$$
$$(\rho u)_t + (\rho u^2 + p)_x = 0.$$

Conservation form:

$$U_t + F(U)_x = 0.$$

Characteristic speeds:

$$u \pm c, \quad c^2 = p'(\rho).$$

Wave speed depends on state.

Structural Summary of Section 3.1

- Burgers: nonlinear scalar conservation law.
- Wave equation: linear hyperbolic system.
- p-system: nonlinear elasticity.
- Gas dynamics: physical origin of hyperbolic systems.
- Characteristic speeds $=$ eigenvalues of flux Jacobian.

This prepares for the formal theory of characteristics in Section 3.2.